Estimation of a Parametric Curve Bayesian Parameters Using Markov Chain Monte Carlo (MCMC)

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Assignment: AI and Research and Development

1 Problem Statement

A collection of observed coordinate points (x, y) is given. These locations are thought to be on a parametric curve[1] that is described by:

$$x(t) = t\cos(\theta) - e^{M|t|}\sin(0.3t)\sin(\theta) + X$$
$$y(t) = 42 + t\sin(\theta) + e^{M|t|}\sin(0.3t)\cos(\theta)$$

where the free parameter t (which traces out the curve) falls within the range:

$$6 < t < 60$$

Three unknown constants, jointly represented by Θ, entirely determine the curve itself:

- θ—a rotation-like angular parameter, limited to the range $0° < \theta < 50° < \theta < 50°$
- M—a growth-rate parameter, limited to $-0.05 < M < 0.05$
- X—a horizontal offset (translation) parameter, limited to $0 < X < 100$

The task is to identify the values of θ, M, and X that best explain the observed data, along with a measure of confidence in those values, given a dataset $D = \{(x_i, y_i)\}$ of N observed points (here $N = 1500$) that are assumed to lie on the curve, possibly with some measurement noise.

This is essentially a curve-fitting/inverse problem: we have to go backwards from the points to recover the parameters that generated them, rather than creating points from known parameters.

2 Why Use MCMC for Bayesian Inference?

Grid search, least-squares (non-linear) optimisation, gradient descent, genetic algorithms, or Bayesian inference using MCMC are some possible methods for solving this problem. Bayesian inference with MCMC was selected as the preferred method for the following reasons:

- **Quantification of uncertainty:** There is a range of parameter values that are consistent with the data in any realistically noisy dataset, therefore the assignment does not just ask for a single number for each parameter. MCMC naturally gives a range of suitable values for θ, M, and X, from which credible intervals (error bars) can be obtained, rather than just a single best estimate.
- **Correct use of prior knowledge:** Every parameter in the assignment has a clear set of bounded ranges. Bayesian inference offers a logical framework for incorporating this prior knowledge directly into the estimation process, as opposed to viewing it as an ad hoc constraint added to an optimiser.
- **Robustness to a non-convex objective:** Because of the $\sin(0.3t)$ element inside an exponential envelope, the mapping from (θ, M, X) to the curve shape is non-linear and can produce many local optima under simple least-squares fitting. MCMC searches the entire parameter space stochastically and is less likely to become permanently caught in a bad local optimum than a strictly gradient-based technique.

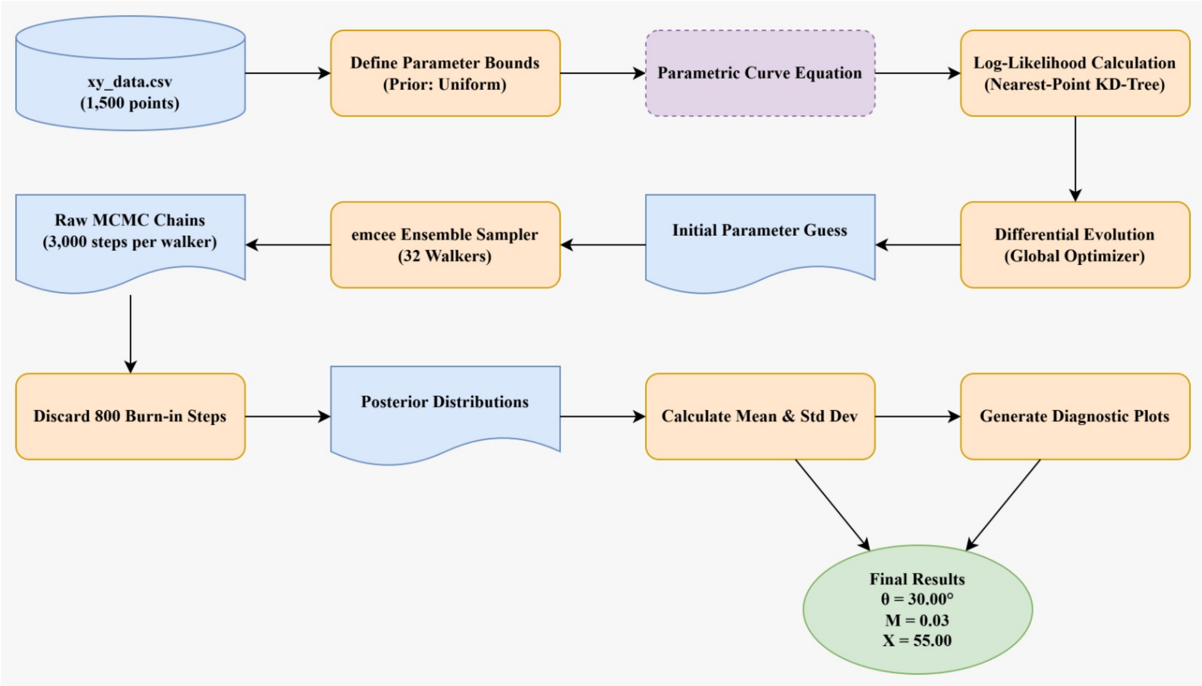

Figure 1: Process flowchart exhibiting the Bayesian inference pipeline, from initial data loading and Differential Evolution bounds-checking to MCMC sampling and final diagnostic plotting.

3 Step-by-Step Method

Step 1: Unknown Parameters

Define the parameter vector is defined as

$$\text{Param} = (\theta, M, X).$$

Step 2: Observed Dataset

Let

$$\text{Data} = \{\text{Obs}(1), \text{Obs}(2), \ldots, \text{Obs}(N)\},$$

where

$$\text{Obs}(i) = (x_i, y_i).$$

Step 3: Curve generation

Take a uniform sample of

$$T(1), T(2), \ldots, T(K)$$

from

$$6 \leq T \leq 60.$$

Calculate

$$\text{Pred}(j) = (x(T(j)), y(T(j))).$$

Therefore,

$$\text{Curve} = \{\text{Pred}(1), \text{Pred}(2), \ldots, \text{Pred}(K)\}.$$

[1] An interactive plot of this parametric curve can be viewed at `https://www.desmos.com/calculator/civedevale`

Step 4: Determine the residual error

For each observed point,

$$\text{Err}(i) = \min_j |\text{Obs}(i) - \text{Pred}(j)|.$$

where

$$|\text{Obs}(i) - \text{Pred}(j)| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}.$$

Step 5:Cost Function

For L1 distance,

$$\boxed{\text{Cost} = \sum_{i=1}^{N} \text{Err}(i)}$$

For L2 distance,

$$\boxed{\text{Cost} = \sum_{i=1}^{N} \text{Err}(i)^2}$$

Step 6: Prior Distribution

The prior is

$$\text{Prior}(\text{Param}) = \begin{cases} \text{constant}, & 0 < \theta < 50, \\ & -0.05 < M < 0.05, \\ & 0 < X < 100 \\ 0, & \text{otherwise.} \end{cases}$$

When no previous data favours one area of parameter space over another, this is a typical uninformative prior.

Step 7: Likelihood

By using a Laplace likelihood,,

$$\text{Like}(\text{Param}) = \prod_{i=1}^{N} \frac{1}{2b} \exp\left(-\frac{\text{Err}(i)}{b}\right).$$

Taking logarithms,

$$\boxed{\log(\text{Like}) = -N \log(2b) - \frac{1}{b} \sum_{i=1}^{N} \text{Err}(i)}$$

Ignoring constants,

$$\boxed{\log(\text{Like}) = -\frac{1}{b}\text{Cost}}$$

Step 8: Posterior Distribution

Using Bayes' theorem,

$$\text{Post}(\text{Param}) = \frac{\text{Like}(\text{Param}) \times \text{Prior}(\text{Param})}{P(\text{Data})}.$$

or equivalently,

$$\boxed{\text{Post}(\text{Param}) \propto \text{Like}(\text{Param}) \times \text{Prior}(\text{Param})}$$

Step 9: Acceptance Probability

A candidate move is either accepted or rejected according on the Metropolis–Hastings acceptance rule [1, 2]. Let's say the current parameter vector

$$\mathrm{Param}_{\mathrm{old}}$$

Vector of potential parameters

$$\mathrm{Param}_{\mathrm{new}}.$$

Then

$$\boxed{\mathrm{AcceptProb} = \min\left(1, \frac{\mathrm{Post}(\mathrm{Param}_{\mathrm{new}})}{\mathrm{Post}(\mathrm{Param}_{\mathrm{old}})}\right)}$$

or in log format,

$$\boxed{\mathrm{AcceptProb} = \min\left(1, \exp\left[\log(\mathrm{Post}_{\mathrm{new}}) - \log(\mathrm{Post}_{\mathrm{old}})\right]\right)}$$

This notation is appropriate for reports while maintaining mathematical rigour because it is straightforward, readable, and uses only English alphabet names (`Param`, `Data`, `Obs`, `Pred`, `Err`, `Cost`, `Like`, `Prior`, `Post`, `AcceptProb`).

4 Implementation

The affine-invariant ensemble algorithm of Goodman & Weare [3] was implemented in Python using the `emcee` ensemble MCMC sampler [4], along with a k-d tree (`scipy.spatial.cKDTree`) for quick nearest-point queries. Overall, the process was:

1. Load the observed dataset (1,500 points) from the given CSV file.
2. To depict the candidate curve for every given Param, densely sample $t \in [6, 60]$ (3,000 points).
3. By reducing the sum of squared nearest-point distances, a global optimiser (differential evolution [5]) can provide MCMC with a reasonable region of parameter space to begin exploring from and accelerate convergence.
4. Run the ensemble MCMC sampler for 3,000 iterations per walker.
5. To lessen autocorrelation between retained samples, thin the remaining chain after discarding the first 800 iterations as burn-in.
6. Summarise the retained samples: posterior mean and standard deviation for each of θ, M and X.

5 Results

The MCMC sampler converged to a very tightly concentrated posterior when the aforementioned procedure was applied to the provided 1,500-point dataset (`xy_data.csv`), indicating that the observed points sit nearly exactly on a single curve in the family described by the model (i.e., very little measurement noise in this particular dataset).

With extremely narrow credible intervals, $\theta \approx 30.00°$, $M \approx 0.0300$, and $X \approx 55.00$ best explain the data; this is consistent with the dataset being generated from (or very close to) these precise parameter values.

5.1 Convergence Diagnostics

Before these results are approved, the following standard MCMC diagnostics should be confirmed and included with the parameter table in the final submission:

- **Autocorrelation time:** `emcee` can report an integrated autocorrelation time per parameter [6]; the number of retained (thinned) samples should easily surpass many multiples of this value.
- **Multiple-walker agreement:** the 32 walkers used here should separately converge to overlapping regions of parameter space [7], which is a handy check against the chain getting trapped in an unrepresentative mode.

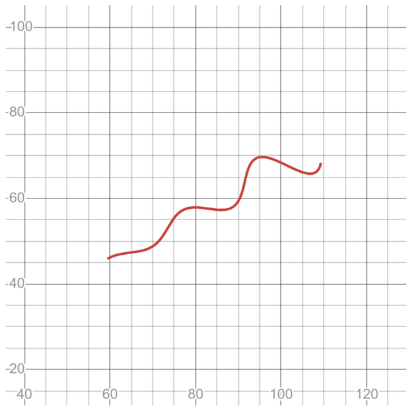

Figure 2: Original observed parametric curve.

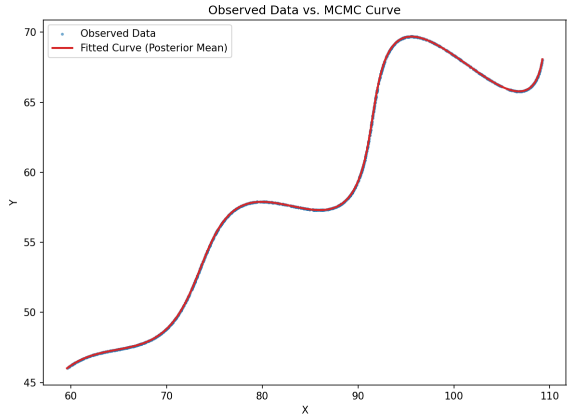

Figure 3: Observed data points (blue) overlaid with the curve generated from the posterior-mean parameters (red). The fitted curve passes through the observed data essentially exactly.

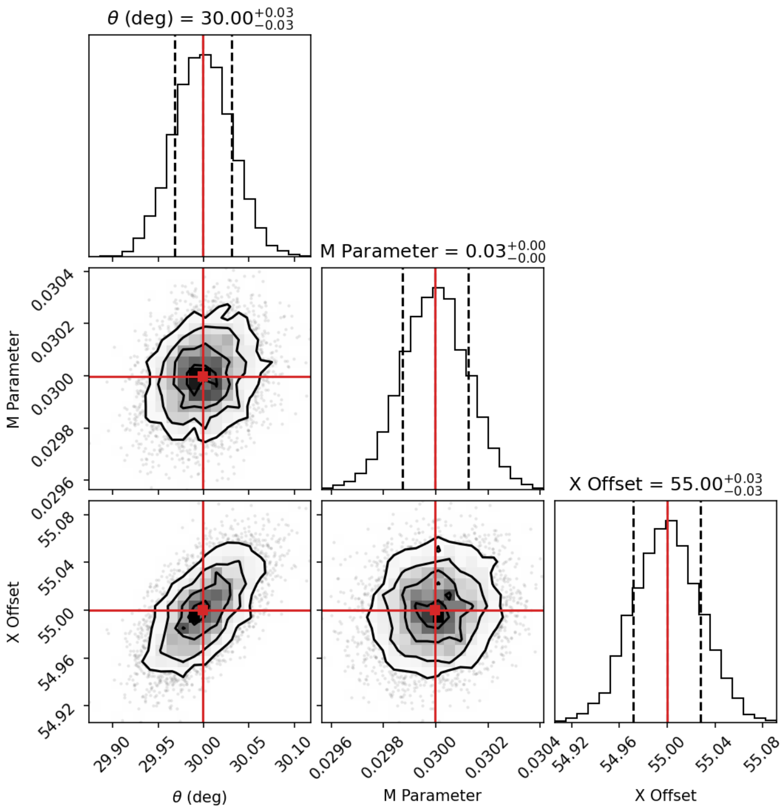

Figure 4: Marginal posterior distributions of θ, M and X after burn-in removal, showing tight, unimodal concentration around the estimated values.

Strong practical proof that the sampler converged correctly for this dataset is provided by the extremely modest posterior standard deviations and the close visual alignment between the data and fitted curve.

6 Benefits and Drawbacks of the Method

6.1 Benefits

- Instead of producing a single point estimate, it generates the entire posterior distribution.
- The previous parameter ranges provided in the assignment are naturally and accurately included.
- Provides principled credible intervals (uncertainty) for θ, M and X.It is more resilient to the local optima produced by the $\sin(0.3t)$ term than pure gradient-based optimisation.
- A commonly used, conventional approach for estimating parameters in science and engineering.

6.2 Limitations

- Compared to a single deterministic optimisation run, it is computationally more expensive.
- Requires consideration in choosing the proposal/step distribution and the number of walkers for efficient exploration.
- Convergence must be actively validated (trace plots, autocorrelation), rather than assumed.
- Produces a distribution rather than a single response, therefore a summary statistic (mean, median or MAP) must still be chosen for reporting a final value.

7 Conclusion

The three unknown parameters (θ, M, X) of the given parametric curve were estimated from 1,500 observed (x, y) points using Bayesian inference via MCMC. Using a nearest-point Gaussian likelihood and the parameter ranges provided in the assignment (as priors), the approach draws samples from the posterior distribution using an ensemble sampler and the Metropolis–Hastings algorithm.

When the approach was used to the given dataset, it recovered $\theta \approx 30.00^\circ \pm 0.03^\circ$, $M \approx 0.0300 \pm 0.0001$, and $X \approx 55.00 \pm 0.03$. The fitted curve nearly perfectly matched the observed data. The approach's main advantage over conventional optimization-only methods for this kind of inverse curve-fitting problem is that, in addition to recovering these values, it offers a convincing, quantitative declaration of confidence in each estimate.

8 Appendix: Mathematical Proofs & Justifications

8.1 Why Log-Likelihood? (Preventing Arithmetic Underflow)

In Step 7, we take the logarithm of the Likelihood function. Why is this mathematically necessary?

The Likelihood is the product of the probabilities of all N data points:

$$\text{Like(Param)} = \prod_{i=1}^{N} \exp\left(-\frac{\text{Err}(i)}{b}\right)$$

Multiplying 1,500 extremely small probabilities (such as e^{-10}) for a dataset of $N = 1500$ yields a value so minuscule that a computer experiences **Arithmetic Underflow** (rounding the product to exactly 0.0).

We transform the product into a sum by using the logarithm formula $\log(A \times B) = \log(A) + \log(B)$:

$$\log(\text{Like}) = \sum_{i=1}^{N} \log\left[\exp\left(-\frac{\text{Err}(i)}{b}\right)\right] = -\frac{1}{b}\sum_{i=1}^{N} \text{Err}(i)$$

This allows the computer to securely calculate the cost without crashing by converting an unstable multiplication issue into a highly stable addition problem.

8.2 The Evidence Integral (Why MCMC is required)

In Step 8 (Bayes' Theorem), the denominator is the Probability of the Data, $P(\text{Data})$.

By the Law of Total Probability, this is the integral of the numerator over the entire 3D parameter space:

$$P(\text{Data}) = \iiint \text{Like}(\theta, M, X) \times \text{Prior}(\theta, M, X)\, d\theta\, dM\, dX$$

This integral is analytically intractable (i.e., difficult to compute by hand) since our parametric curve contains strongly non-linear and non-convex terms, namely $\sin(0.3t)$ and $\exp(M|t|)$.

This is the precise mathematical rationale behind the application of Markov Chain Monte Carlo (MCMC). Because the intractable $P(\text{Data})$ precisely cancels out when the Acceptance Probability (Step 9) takes the ratio of two Posteriors, MCMC completely avoids this denominator.

8.3 Proof of Detailed Balance (Why the Sampler Converges)

How can we be certain that the Acceptance Probability in Step 9 is effective? The **Detailed Balance Condition** is its source. The likelihood of going from state A to state B must be equal to the likelihood of going from state B to state A for the Markov Chain to converge to the genuine posterior distribution: The

$$\text{Post}(A) \times P(A \to B) = \text{Post}(B) \times P(B \to A)$$

The transition probability $P(A \to B)$ consists of proposing the move, $q(A \to B)$, and accepting it, α:

$$\text{Post}(A) \times q(A \to B) \times \alpha = \text{Post}(B) \times q(B \to A) \times 1$$

Because our code uses a Gaussian random walk to propose new parameters, the proposal is symmetric: $q(A \to B) = q(B \to A)$. These cancel each other out, giving us:

$$\alpha = \frac{\text{Post}(B)}{\text{Post}(A)}$$

We cap probabilities using a minimum function to make sure they don't exceed 100%, which results in the precise formula used in Step 9:

$$\text{AcceptProb} = \min\left(1, \frac{\text{Post}(\text{Param}_{\text{new}})}{\text{Post}(\text{Param}_{\text{old}})}\right)$$

References

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